

CS & IT ENGINEERING



THEORY OF COMPUTATION

Pushdown Automata

Lecture – 03



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Recap of Previous Lecture



Topic

PDA Construction
?????

NPDA Construction

DPDA Construction



Topics to be Covered



Topic

Push down automata

Topic

??

Context free language
Detection

Topic

??

Topic

??

Closure properties



Topic : Pushdown Automata

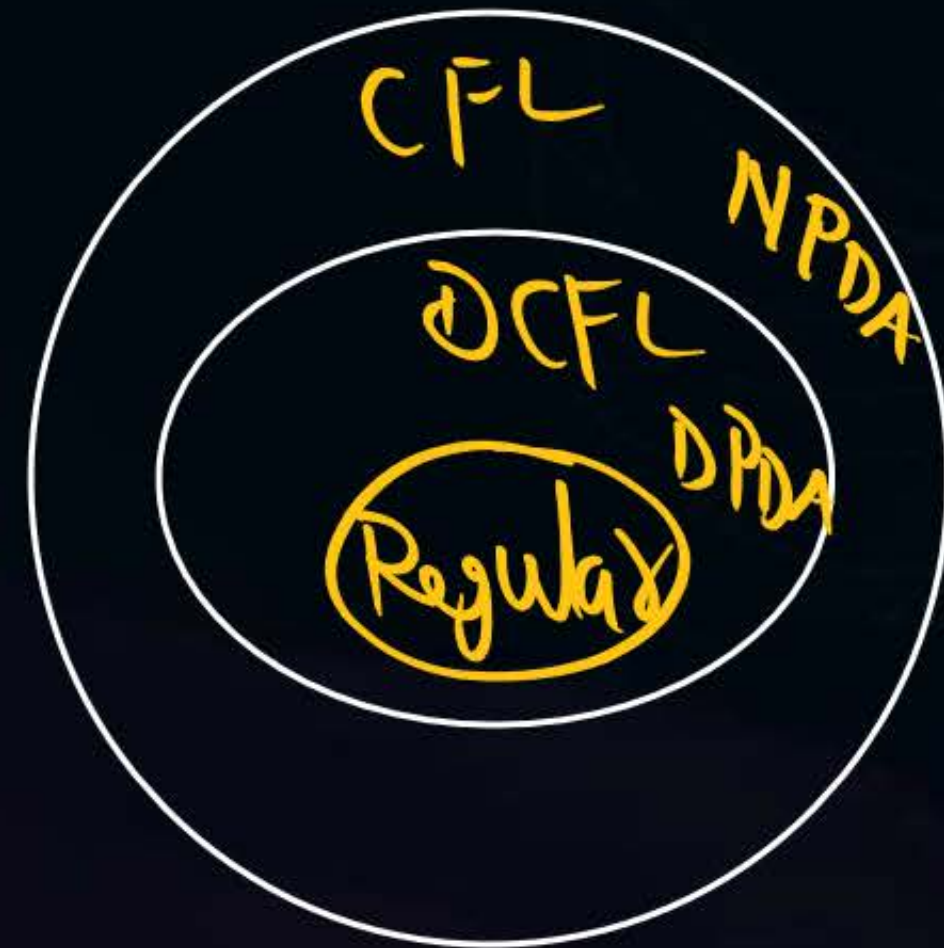
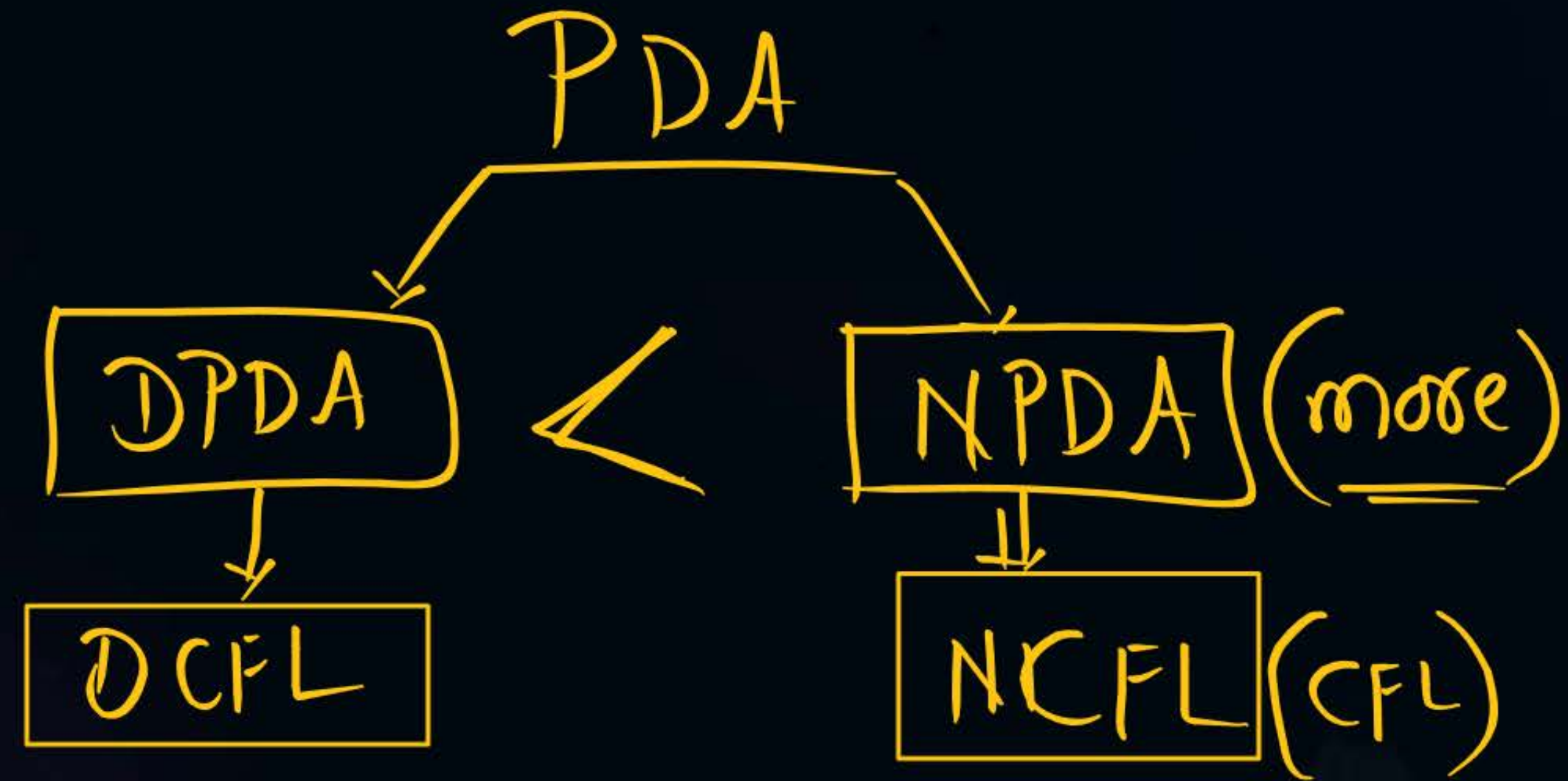
a a a a - -

$$L = \underbrace{\{a^n b^{2n}\}}_{xx} \cup \underbrace{\{a^n b^{3n}\}}_{xxx}$$

X DPDA?
NPDA

$$(q_0, a) = (q_1, xx) \text{ (or) } (q_1, xxx)$$

CFL but not DCFL



$L = \{WWR\}$

DPDA $\times$

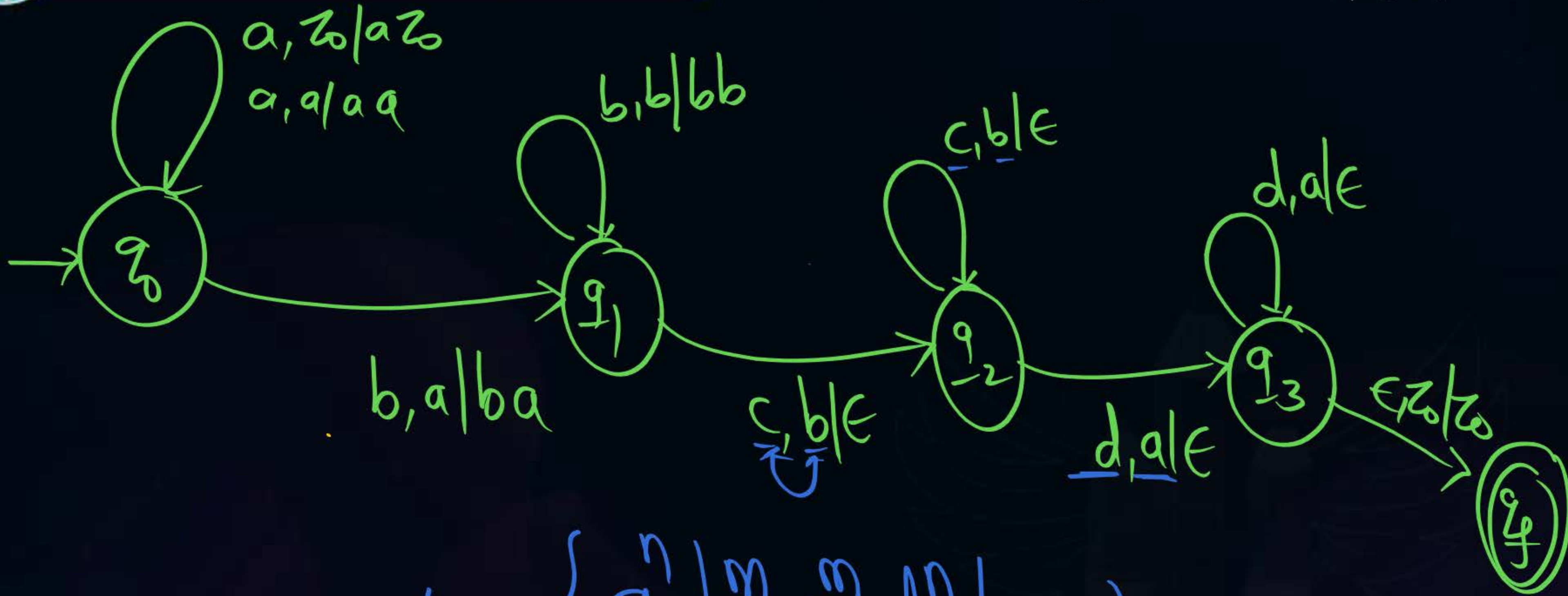
CFL but not DCFL

NPDA $\checkmark$



Topic : Pushdown Automata

Identify language of given PDA?

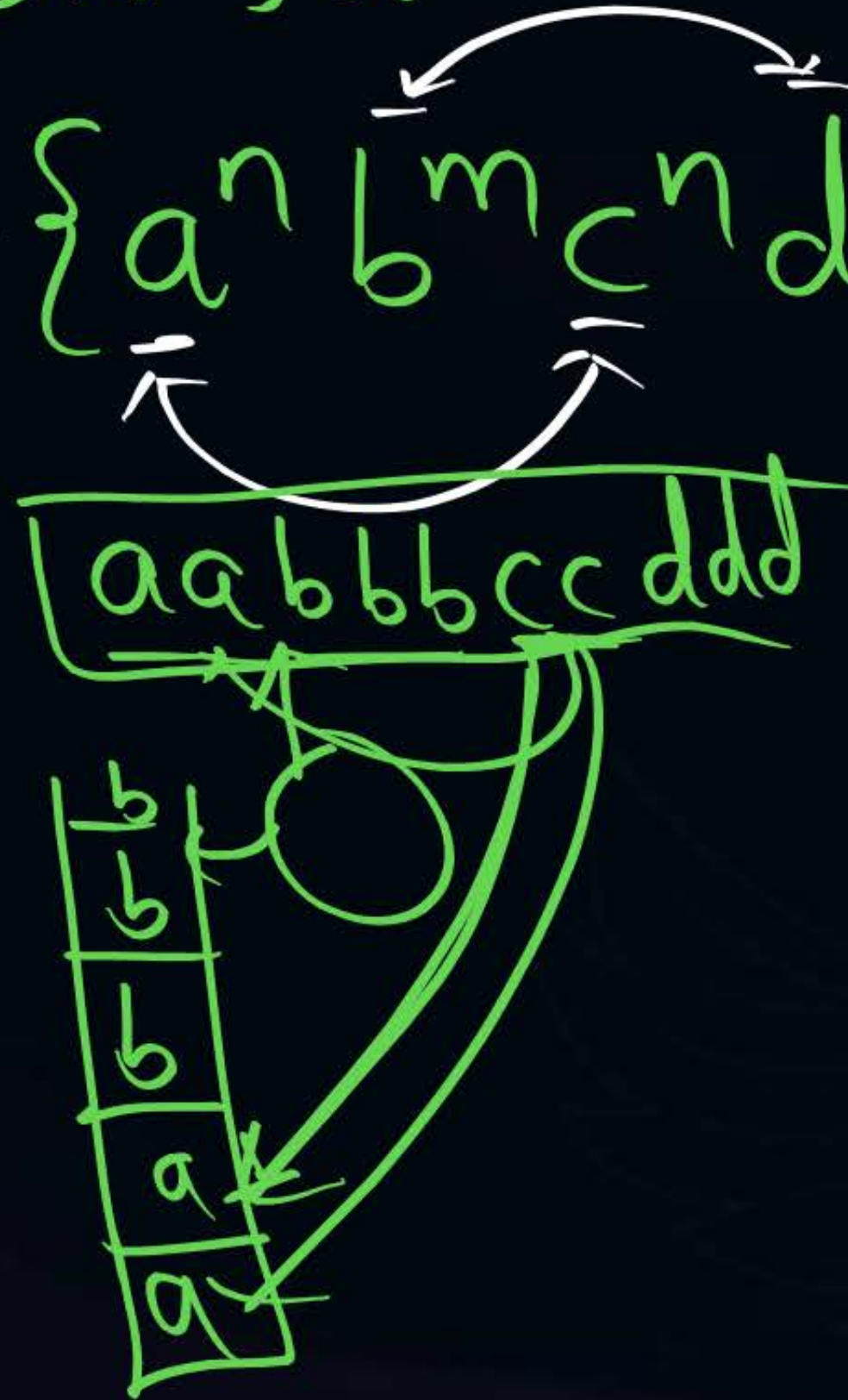


$$L = \{ a^n b^m c^m d^n \mid n, m \geq 1 \}$$

(Q) Construct PDA for

$$L = \{a^n b^m c^n d^m \mid n, m \geq 1\}$$

PDA not possible



Logic

$a's \rightarrow$

$b's \rightarrow$

$c's \rightarrow$

$$L_1 = \{a^m \overrightarrow{b^n} \overrightarrow{c^n} \overrightarrow{d^m} \mid n, m \geq 1\} \text{ CFL}$$

$$L_2 = \{a^n \overrightarrow{b^m} \overrightarrow{c^n} \overrightarrow{d^m} \mid n, m \geq 1\} \text{ non CFL}$$

(Q) Construct PDA for

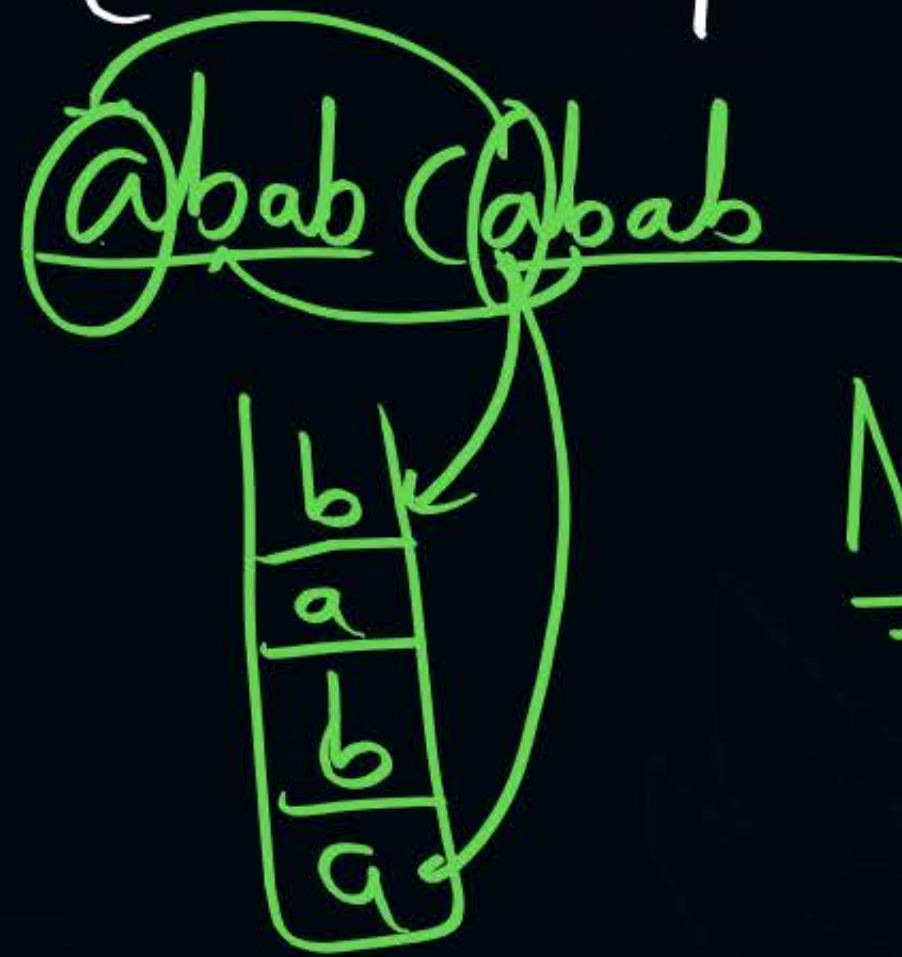
$$L = \{ w c w / w \in (a+b)^* \}$$

aa c bb



PDA
not possible

abab c abab



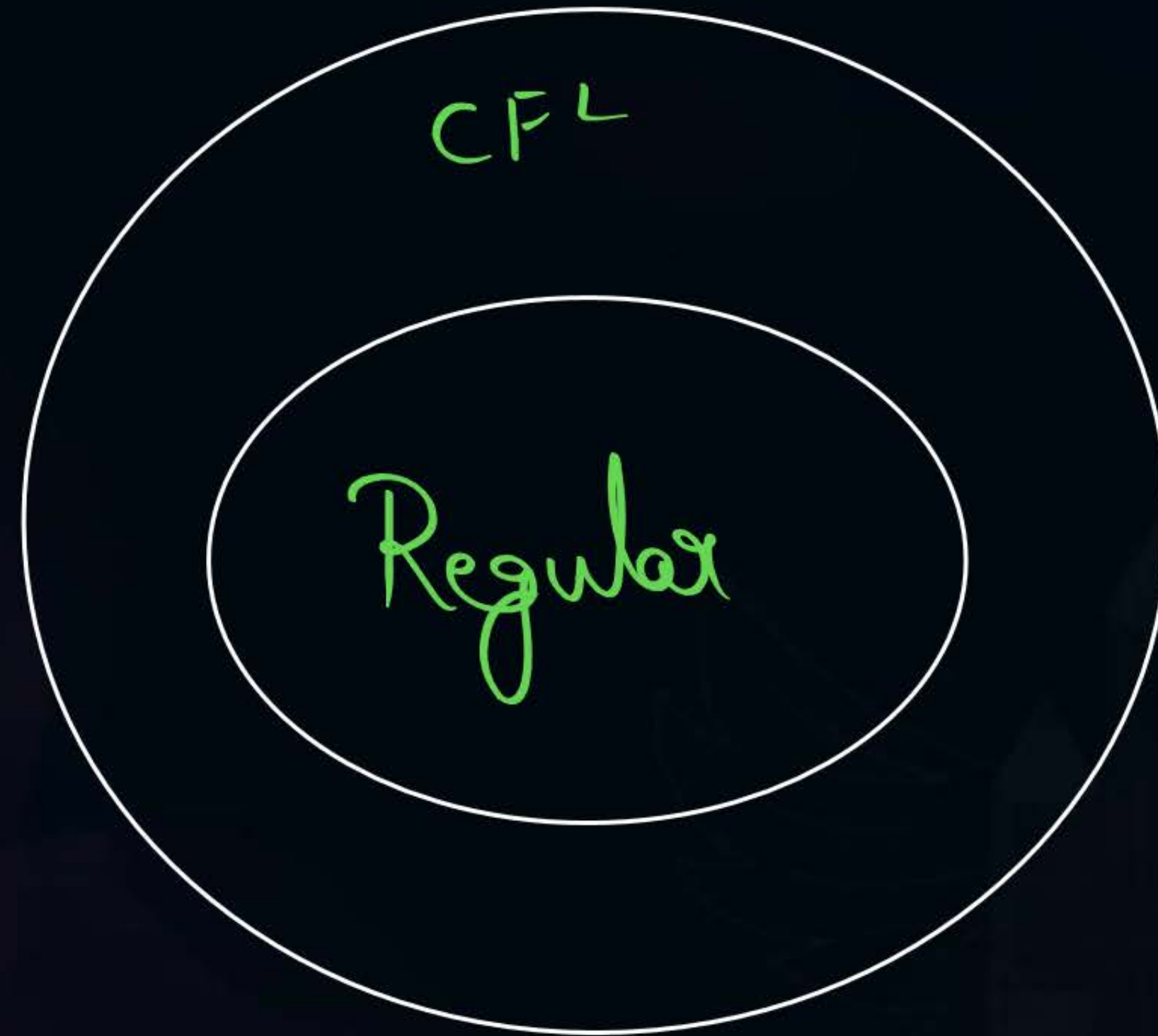
Non CFL

Logic

$L = \{a^n b^n c^n \mid n \geq 1\}$ PDA not possible

a a b b c c







Topic : Context Free Language



Which of the following language are

1. CLF and Regular
2. CFL but not regular
3. Non CFL



Topic : Context Free Language

1. $L = \{a^n b^n c^n / n \leq 10\} \rightarrow \text{finite} \rightarrow \text{Regular} \rightarrow \text{CFL}$

2. $L = \{a^n b^n c^m / n \neq m\} \Rightarrow \text{Non CFL}$

3. $L = \{a^n b^m c^n / n > m, n, m \leq 1000\} \rightarrow \text{finite} \rightarrow \text{Reg} \rightarrow \text{CFL}$

4. $\{L = \{a^n b^m / n - m = 4\} \rightarrow \text{CFL}$
 $n = m + 4$

$$\{a^{4m}b^m\}$$

⑤ $L = \{a^n b^m / n/m = 4\} \rightarrow CFL$

⑥ $L = \{a^n b^m / n = 2m + 1\} = \{a^{2m+1} b^m\} \rightarrow \{ \underline{CFL} \text{ but not } \underline{\underline{regular}} \}$

⑦ $L = \{a^n b^m / n \neq m\} \rightarrow CFL$

8. $L = \{a^n b^m / n \neq 2m\} \rightarrow CFL$

⑨ $L = \{a^n b^m / n = m^2\} \rightarrow \underline{\underline{NOT CFL}}$

$\{a^{m^2} b^m\}$



Topic : Context Free Language

⑩ $\{a^n b^n / n \geq 13\} \rightarrow \underline{\text{non}} \underline{\text{CFL}}$

⑪ $L = \{a^n b^m / n \leq m\} \rightarrow \text{CFL}$

⑫ $\{a^n b^m c^{n+m} / n, m \geq 1\} \rightarrow \text{CFL}$

push push pop

13. $L = \{L = a^n b^{n+m} / n, m = 1\}$

14. $L = \{a^{m^2} b^{n^3} c^{k^5} / m, n, k \geq 1\} \rightarrow \text{non CFL}$

15. $L = \{a^{3^n} b^{5^k} c^{2^\ell} / n, k, \ell \geq 1\} \rightarrow \underline{\text{non CFL}}$

$\{a^i b^{i+k} c^k\} = \{ \underbrace{a^i b^i}_{(a^i b^i)} \underbrace{b^k c^k}_{(b^k c^k)} \}$

16. $L = \{a^i b^j c^k / j = i + k\} \Rightarrow \underline{\text{CFL}}$ but not regular

18. $L = \{a^i b^j c^k \mid i > j \text{ and } j > k\} \rightarrow \text{non CFL}$

17. $L = \{a^i b^j c^k / i > j \text{ (or) } j < k\} \rightarrow \text{CFL}$



Topic : Context Free Language

$a^n b^m c^m d^n$

18. $L = \{a^i b^j c^k / i > j > k\}$

19. $L = \{a^i b^j c^k / j = \max(i, k)\} \rightarrow \text{Non CFL}$

20. $L = \{a^i b^j c^k / j = i^2 + k^2\} \rightarrow \text{Non CFL}$

21. $L = \{a^i b^j c^k / d^\ell / i = \ell \text{ (and) } j = k\} \text{ CFL}$

22. $L = \{a^i b^j c^k / d^\ell / i = k \text{ (and) } j = \ell\} \rightarrow \underline{\underline{\text{Non CFL}}}$

$$23. \quad L = \{a^i b^j c^k / d^\ell / \underline{i = k} \text{ (or) } \underline{j = \ell}\} \rightarrow CFL$$

$$24. \quad L = \{a^i b^j c^k / d^\ell / i = 2k \text{ (or) } j \neq 5\ell\} \rightarrow CFL$$

$$25. \quad L = \{a^i b^j c^k / d^\ell / i + j = k + \ell\}$$

$$26. \quad L = \{a^i b^j c^k / d^\ell / i = 4\ell \text{ and } j = 3\ell\}$$



Topic : Context Free Language

27. $L = \{a^i b^j / (i + j) \bmod 5 = 0\}$

28. $\{a^{2^n} / n \geq 1\} \rightarrow \text{not CFL}$

29. $\{a^{n^2} / n \geq 1\} \rightarrow \text{not CFL}$

30. $L = \{1^{2n+1} / n \geq 1\} \rightarrow \text{CFL}$

31. $L = \{a^p / p \text{ is prime number}\} \rightarrow \text{not CFL}$

32. $L = \{a^k / k \text{ is odd number}\} \rightarrow \text{CFL}$

$$(38) L = \{ \underline{w} \underline{w} / w \in (a+b)^* \} \rightarrow \text{non CFL}$$

$$(39) L = \{ \underbrace{w}_{\bar{w}} \underbrace{w}_{\bar{w}} \underbrace{w}_{\bar{w}} / w \in \{a, b\}^* \} \rightarrow \text{non CFL}$$

$$(40) L = \{ x / x \in \{a, b, c\}^* \mid n_a(x) = n_b(x) = n_c(x) \} \rightarrow \text{non CFL}$$

$$(41) L = \{ x / x \in \{a, b, c\}^* \mid n_a(x) = n_b(x) + n_c(x) \} \rightarrow \text{non CFL}$$

$$(42) L = \{ x / x \in \{a, b, c\}^* \mid n_a(x) = n_b^2(x) + n_c^2(x) \} \rightarrow \underline{\text{non CFL}}$$



Topic : Context Free Language

43. $L = \{x/x \in \{a, b\}^* \mid n_a(x) \bmod 5 \text{ and } n_b(x) \bmod 4 = 0\}$

44. $L = \{a^n b^{2n} c^{3n} \mid n \geq 1\}$

45. $L = \{a^n b^n c a^m b^m \mid n, m \geq 0\}$

46. $L = \{a^n b^m c^k \mid n \neq m \text{ or } m \neq k\}$



Topic : Context Free Language

- 47. ^{CFL} Set of all odd length palindrome string of Hindi language
- 48. ^{CFL} Set of all even length palindrome string of are language
- 49. ^{qil} ⁽⁽⁽⁾⁾⁾ Set of all balanced parenthesis
- 50. ~~Set of all lexical error produced by compiler~~

Q

Consider the following language over the alphabet $\Sigma = \{a, b, c\}$.

Let $L_1 = \{a^n b^n c^m \mid m, n \geq 0\}$ and $\rightarrow \text{CFL}$

$L_2 = \{a^m b^n c^n \mid m, n \geq 0\}$. $\rightarrow \text{CFL}$

Which of the following are context-free languages?

I. $L_1 \cup L_2 \rightarrow \text{CFL}$

II. $L_1 \cap L_2 \rightarrow \times$

[2017(Set-1): 2 Marks]

A

I only

B

II only

C

I and II

D

Neither I nor II

Q

Let L_1, L_2 be any two context-free languages and R be any regular language. Then which of the following is/are CORRECT?

- I. $L_1 \cup L_2$ is context-free $\rightarrow$ CFL
- II. $\bar{L}_1$ is context-free $\rightarrow$ false
- III. $L_1 - R$ is context-free $\Rightarrow$ CFL - reg $\Rightarrow$ CFL $\wedge$ reg^C = CFL $\wedge$ reg - CFL
- IV. $L_1 \cap L_2$ is context-free $\Rightarrow$

[2017(Set-2): 1 Marks]

- A I, II and IV only
- B I and III only
- C II and IV only
- D I only



Topic : CONTEXTFREE LANGUAGE

closure Properties of DCFL and CFL

<u>Operation</u>	<u>CFL</u>	<u>DCFL</u>
① Union op	✓	✗
② Concatenation op	✓	✗
③ Complement op	✗	✓
④ Intersection op	✗	✗
⑤ ^{CFL ∩ Reg} Intersection with Regular	^{CFL} ✓	^{DCFL} ✓
⑥ Union with Regular	✓	✓

Operation	<u>CFL</u>	<u>DCFL</u>
⑦ Kleene closure	✓	×
⑧ Positive closure	✓	×
⑨ Difference op $[L_1 - L_2 = L_1 \cap L_2^c]$	×	×
⑩ Subset op	×	×
⑪ L -reg $[L \cap \text{reg}]$ CFL-reg	✓	✓
⑫ regular $\text{---} L$ $[reg \cap L]$ CFL	×	✓

CFL-reg
C

<u>Operation</u>	<u>CFL</u>	<u>DCFL</u>
⑬ <u>Substitution op</u>	✓	✗
⑭ <u>Homomorphism op</u>	✓	✗
⑮ <u>Inverse Homomorphism op</u>	✓	✓



2 mins Summary



Topic

One

Topic

Two

Topic

Three

Topic

Four

Topic

Five



THANK - YOU